Muzammal Naseer

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Education

2017 - 2020	**PhD** - Engineering & Computer Science Australian National University (ANU) Thesis: "Analysing & Improving Neural Network's Robustness against Adversarial Perturbations."
2011 - 2013	**Master** of Science in Electrical Engineering King Fahd University of Petroleum & Minerals (KFUPM) Thesis: "The Design of Finite Impulse Response (FIR) Wavefield Extrapolation Filters." GPA: **3.75/4.0**, *First Class Honours.*
2006 - 2010	**Bachelor** of Science in Electrical Engineering University of the Punjab (PU) Thesis: "Spectrum Monitoring & Analysis of Interference in EGSM Networks." GPA: **3.89/4.0**, *First Class Honours, Gold Medalist.*

Work Experience

Sept. 2020 - Present	*Research Associate* **Mohamed bin Zayed University of Artificial Intelligence - MBZUAI**, UAE Developing novel algorithms for adversarial attacks and defenses, few-shot learning and domain generalization. I am also supervising MS/Ph.D. students and research assistants.
Sept. 2018 - Mar. 2020	*Research Assistant* **Inception Institute of Artificial Intelligence - IIAI**, UAE Developed novel adversarial attack and defense algorithms for deep neural networks.
Jan. - Jul. 2018	*Research Assistant* **Data61, CSIRO**, Canberra, Australia Developed new black-box attack methods for transferable adversarial perturbations for neural networks based vision systems.
Sept. 2014 - June. 2017	*Lecturer (Full-Time)* **Department of Electrical Engineering - subcampus of KFUPM**, KSA I joined subcampus of KFUPM, now known as "University of Hafr Al-Batin" as a full-time lecturer to teach variety of Electrical Engineering courses.
Jan. - May 2013	*Teaching Assistant* **Department of Electrical Engineering, KFUPM**, KSA Assisted in Control Theory and Signal Processing courses for the period of one semester.
May - June 2009	*Research Intern* **Frequency Allocation Board (FAB)**, Pakistan Worked with FAB team to develop the direction finding system for RF Interferers.

Research Interests

I am intrigued by brittleness of deep neural networks in (non) i.i.d settings, thus I am focused on understanding and improving out-of-distribution generalization of these models. Currently, I am developing new algorithms for "Adversarial Attack and Defense" methods, "Attention based Modeling", and "Few-shot" learning. My research includes applications to classification, segmentation, object detection and visual tracking.

Publications

Google Scholar

M. Naseer, S. Khan, M. Hayat, F. Shahbaz, MH Yang. *"Intriguing Properties of Vision Transformers"*, Advances in Neural Information Processing Systems (*NeurIPS-2021–Spotlight*) [Paper, code]

M. Naseer, S. Khan, M. Hayat, F. Shahbaz, F. Porikli. *"On Generating Target Transferable Perturbations"*, International Conference on Computer Vision (*ICCV-2021*) [Paper, code]

K. Ranasinghe, **M. Naseer**, S. Khan, M. Hayat, F. Shahbaz. *"Orthogonal Projection Loss"*, International Conference on Computer Vision (*ICCV-2021*) [Paper, code]

M. Naseer, S. Khan, M. Hayat, F. Shahbaz, F. Porikli. *"A Self-supervised Approach for Adversarial Robustness"*, IEEE Conference on Computer Vision and Pattern Recognition (*CVPR-2020–Oral*) [paper, Presentation, Poster, code].

M. Naseer, S. Khan, H. Khan, F. Shahbaz, F. Porikli. *"Cross-Domain Transferability of Adversarial Perturbations"*, Advances in Neural Information Processing Systems (*NeurIPS'2019*) [paper, poster, code]

M. Naseer, S. Khan, F. Porikli. *"Local Gradients Smoothing: Defense against localized adversarial attacks"*, Winter Conference on Applications of Computer Vision (*WACV'2019*) [paper]

Publications

M. Naseer, S. Khan, F. Porikli. *"Indoor Scene Understanding in 2.5/3D for Autonomous Agents: A Survey "*, Multidisciplinary Open Access Journal (*IEEE Access'2019*) [paper]

M. Naseer, W. Mousa. *"Linear complementarity problem: A novel approach to design finite-impulse response wavefield extrapolation filters"*, Society of Exploration Geophysicists. (*GEOPHYSICS'2014*) [paper]

M. Naseer, W. Mousa, T. Al-Naffouri, A. Al-Shuhail. *"One Dimensional Wavefield Extrapolation Filter Design Via L1 Error Approximation"*, SEG Technical Program Expanded Abstracts (*GEOPHYSICS'2015*) [paper]

M. Naseer, K. Ranasinghe, S. Khan, F. Shahbaz, F. Porikli. *"On Improving Adversarial Transferability of Vision Transformers"*, (*arXiv-2021*) [preprint, code]

M. Afham, S. Khan, MH Khan, **M. Naseer**, F. Shahbaz. *"Rich Semantics Improve Few-shot Learning"*, (*arXiv-2021*) [preprint, code]

S. Khan, **M. Naseer**, M. Hayat, SW. Zamir, F. Shahbaz Khan, Mubarak Shah. *"Transformers in vision: A survey"*, (*arXiv-2021*) [preprint]

M. Naseer, S. Khan, M. Hayat, F. Shahbaz, F. Porikli. *"Stylized Adversarial Defense"*, (*arXiv-2020*) [preprint, code]

Honors and Awards

- Dec. 2019 — **Student Travel Award** by Conference on Neural Information Processing Systems (NeurIPS-2019).
- June 2017 — PHD: Awarded **Postgraduate Research Scholarship** by Australian National University for the period of three years.
- June 2017 — PHD: Awarded an ANU HDR **Fee Remission Merit Scholarship** by Australian National University for the period of four years.
- Jan. 2011 — M.Sc: Awarded Fully Funded **Master of Science Scholarship** by Ministry of Higher Education, KSA for the period of two and half years.
- Jan. 2011 — B.Sc: Awarded **Gold Medal** for outstanding performance and first class honors in B.Sc. Electrical Engineering.
- Feb. 2007-09 — B.Sc: Received **University Merit Scholarship** consistently for three years for best student performance by the University of the Punjab.

Computational Skills

- Python — Expert Knowledge- I am extensively using python to build novel machine learning algorithms for the last few years.
- Pytorch — Pytorch is usually my default choice due to its dynamic nature and object-oriented graph design approach.
- Keras — I have used Keras with Tensorflow as back end to build different projects including a defense algorithm against unconstrained adversarial attacks. [paper]
- Matlab — Most of my B.Sc and M.Sc academic research published or otherwise is implemented in Matlab.
- C/C++ — I scored A+ grades in these languages in my B.Sc courses. [Transcript]

References

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Professor at Research School of Engineering, Australian National University (ANU).
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